

Investigation & Applications of Historical Cryptography

Introduction:

Ever since I was young I've always been fascinated by computers, I was mesmerized by the powerful machines we had at our disposal. Following the use of computers in elementary school came discussions regarding online safety. Emphasizing the importance of strong passwords and such, but I never truly understood how passwords and data security worked, or where the building blocks of encryption came from. In the modern age of technology, the importance of encryption became far more apparent, with safeguarding financial transactions or securing online communication, the applications of cryptography are vast and have been evolving for centuries. These discussions about securing our digital footprint sparked curiosity within me as I grew older. I realized that cryptography is not only a shield for us in the online world but also holds captivating puzzles wishing to be solved. Understanding the mechanisms that underpin our online defences and where these mechanisms originated from is what allows us to move forward with future developments while gaining a better grasp on how the world around us works.

Aim:

The aim of this paper will be to successfully complete a detailed exploration regarding a variety of cryptographic methods and ciphers. This paper will both highlight and explain the mathematics behind each of the explored methods, ranging from modular arithmetic to probability and frequency analysis, in order to deepen my own individual understanding as well as the reader's in a concise and easy-to-follow manner. Such a goal will be achieved through a myriad of examples, definitions, and derivations ultimately leading to a holistic comprehension of what cryptography truly is.

Basic Vocabulary of Cryptography:

Encipher: To convert a message to a coded form

Decipher: To convert a coded message to the original form

Plaintext: Original readable message

Ciphertext: Enciphered unreadable message

The Caesar Shift:

The first method of encryption was developed around 100 BC by Julius Caesar for the purposes of war and was aptly named the “Caesar Shift”. It was comprised of simply shifting each letter in a message three letters over. So if he were to send messages to his general, which happened to be intercepted, then they could not be read. (Arizona State University, *What is the Caesar Shift?*)

Shown in mapping notation with:

$$x \rightarrow x + 3$$

Where x is the position of a letter in the alphabet.

However, this will not work for letters later in the alphabet due to the fact that we must stay in the set of the 26 letters. The letter “Y” for example would give us the cipher-text value of 28.

Therefore, applying modulo 26 to the notation satisfies the constraints. Modulo is an operation which outputs the remainder when two numbers are divided, an example is shown below where 28 divided by 26 outputs a remainder of 2.

$$x \rightarrow (x + 3) \bmod 26$$

$$25 \rightarrow (25 + 3) \bmod 26 = 2 \text{ (“B”)}$$

Application of Frequency Analysis:

Although the Caesar Shift provides a base level of security, it can be solved through brute force due to the fact there are only 26 possible shifts of varying magnitude. Using cryptanalysis it can be somewhat reliably decrypted due to the large samples of English literature we have and the frequency of the letters we use. The most used independent letters are “e”, “t”, and “a” with the most frequent bigrams including ‘th’, ‘he’, ‘an’, ‘re’, ‘er’, and finally the most frequent trigrams including ‘the’, ‘and’, ‘ing’. The frequencies of each letter in the English alphabet can be seen in the graph below:

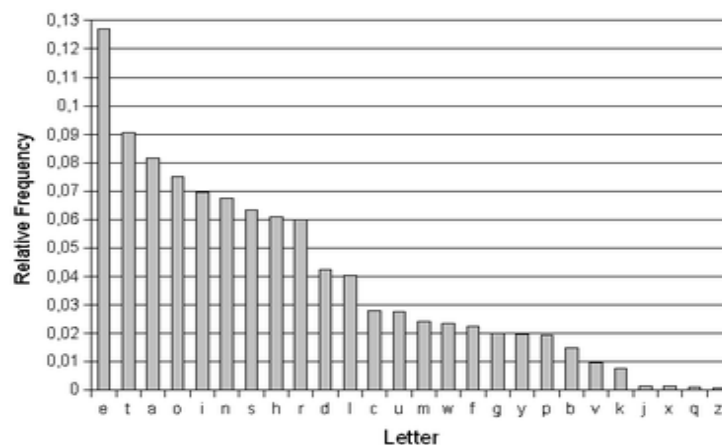


Figure 1 (University of Notre Dame, The Frequency of the Letters of the Alphabet in English)

To demonstrate the plausibility of this form of cryptanalysis I have gotten a friend to write a handful of sentences on an unknown topic and shift each of the letters an unknown amount. The encrypted message comes out to be:

*Gfxpjygfqq nx f lwjfy mtggd fsi N qtaj uqfdnsl bnym rd kwnjsix. Bfyhmns l ymj SGF nx
jsotdfgqj ytt, ymnx djfw N ymnsp ymj ljsajw Szlljyx bnqq jsi zu bnssnsl ymj hmfruntsxmnu.*

Conducting a cryptanalysis based on the frequencies of each letter, I found the following to occur the most:

F: 51, J: 43, S: 36

Since 'e' is the most frequent letter in plain text, we can assume it has been shifted to 'f', or simply a shift of 1 to the right. Scoping in on the first word from the message and reversing this shift we get:

Hgyqkzhgrr

Critically reflecting on this initial failure, we can see that this form of cryptanalysis works better for longer text, but we can repeat this process going down the list of frequencies until we get a meaningful message. We can assume the second most frequent 't' may have been shifted to 'f' which gives us:

Srjbvksrcc

Repeated trial and error in this sense led to my personal conclusion that frequency analysis may be difficult, tedious, and repetitive with smaller texts, and if I were given a larger message then it may prove to be much easier. The third trial is successful, where 'a' has been shifted to 'f', a vector shift of 5 characters right which gives us the fully decrypted message:

Basketball is a great hobby and I love playing with my friends. Watching the NBA is enjoyable too, this year I think the Denver Nuggets will end up winning the championship.

Substitution Ciphers:

The Caesar Shift is a kind of substitution cipher which replaces each letter with a vector transformation, however, more complicated substitution ciphers exist. They include arbitrary mapping and replacement of letters, rather than " $y \rightarrow y + 3$ " in the Caesar Shift, the key to a

substitution cipher may have no pattern at all and will include various arrangements for each letter such as 'B → Q' or 'A → W' as examples (University of Canterbury Computer Science Education Research Group, *Encryption - Substitution Ciphers*). Specifically in the English language, there are 26! possibilities for such rearrangements, where '!' represents factorial meaning 26 is multiplied by every integer less than itself. Finding some of those possibilities would be easier than others due to vast similarities and fixed positions between them, therefore the number of unique permutations (aka derangements) for the English alphabet can be calculated, denoted by !26, which is simply reordering where nothing remains in the same place, this will be explained further using examples below. Finding this number allows us to understand the scope of the encryption we are working with and if it is possible to use software algorithms to help us with decryption in terms of brute force.

Take the set $x \{A, B, C\}$ for example, these three letters can be formatted in 3! different ways:

$$\{A, B, C\} \{B, C, A\} \{C, B, A\} \{A, C, B\} \{B, A, C\} \{C, A, B\}$$

However, we can see similarities between sets where at least one letter is in the same place in both sets. Therefore, through simple inspection, we can see there are two possible derangements not including the original position:

$$\{B, C, A\} \{C, A, B\}$$

But to get to this point through calculation we must look at it from a different perspective. For example, let us say we keep A & B fixed within their position, there would only be 1! way to permute C and we can apply this to fixing the position of B alone, and then no letters:

$$\{A, B, C\}$$

$$\{A, B, C\} \{C, B, A\}$$

$$\{A, B, C\} \{B, C, A\} \{C, B, A\} \{A, C, B\} \{B, A, C\} \{C, A, B\}$$

By doing so we get three different groups of variation, to get rid of any overlapping sets that are the same we can add the first set to the original, and then subtract the second, and extrapolate this pattern if we wanted to (the inclusion-exclusion principle). Applying this to a fraction format and normalizing to our scale would give us:

$$!3 = 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right)$$

$$!3 = 6 \left(1 - 1 + \frac{1}{2} - \frac{1}{6}\right)$$

$$!3 = 2$$

In this case '1' represents the original position where all elements are fixed and the amount of possibilities for rearrangement. The series continues by removing the fixed position of another element and dividing by the permutations for the remaining elements as well as using the inclusion-exclusion principle to get rid of similar sets. Applying this to all possible values gives us the following series:

$$!n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n * \frac{1}{n!}\right)$$

Written in summation as:

$$!n = n! * \sum_{k=0}^n \frac{(-1)^k}{k!}$$

Applied for the English alphabet gives us:

$$!26 = 26! * \sum_{k=0}^{26} \frac{(-1)^k}{k!} \approx 1.5 * 10^{26}$$

A value far too great to be decrypted using brute force or even by software algorithms, even if we had the means of computing one permutation every nanosecond it would still take upwards of approximately 4.8 billion years. Meaning it is virtually impossible.

The Vigenère Cipher

The Vigenère cipher is a method of encryption created by French cryptographer Blaise de Vigenère in the 16th century which employs the use of differing Caesar shifts for each letter in a plaintext. It does so through the use of a secret keyword which is required for decryption, and the Caesar shifts are based on this keyword (Michigan Technological University, *The Vigenère Cipher Encryption and Decryption*). To manually encrypt messages you can use the table below, where the top row represents the letters in the plaintext and the right column represents the letters in the key. For example, to encrypt the letter “J” with the key “S” you would look at where they intersect which is “B” and that is your ciphertext.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Figure 2 (Michigan Technological University, The Vigenère Cipher Encryption and Decryption)

This adds a new layer of complexity due to the fact that frequency analysis cannot be used on the original encrypted message. Leading to the Vigenère cipher remaining unbroken for three centuries, however, there are modern methods we can use to solve it. Once again I have gotten a friend to send me an encrypted message using the table above which I will attempt to crack:

*Yulpo ctu ne t ndeta touik agk ut vhz hxsb chuzeva behwx. B wqrlvzaesk lhcq th wxar ate
ipmnh igt B omvx adixk fo ismy fhzy buetkbyegae ig ate ihet. B iqlblhe moqrx pe shtqtapzg
kpshm mar xcqrrvze.*

The ciphertext was created using an unknown keyword with an unknown length, however, it is possible to decrypt it through the use of probability. Specifically, something known as the Index of Coincidence which is the probability of picking two letters that are the same, comparing this index between the English language and the Caesar-shifted text will allow us to get closer to cracking the code (University of Southampton, Dr. JH Renshaw, *Index of Coincidence*). The variables used to do so are defined as:

n = total number of letters in ciphertext

l = length of keyword

I = Theoretical Index of Coincidence for English

I_v = Theoretical Index of Coincidence for Vigenère cipher

I_e = Experimental Index of Coincidence for ciphertext

By manually counting the letters in the ciphertext I found:

$$n = 158$$

Letter	a	b	c	d	e	f	g	h	i	j	k	l	m
Frequency	.082	.015	.028	.043	.127	.022	.020	.061	.070	.002	.008	.040	.024
Letter	n	o	p	q	r	s	t	u	v	w	x	y	z
Frequency	.067	.075	.019	.001	.060	.063	.091	.028	.010	.023	.001	.020	.001

Figure 3 (Northern Kentucky University, *Cryptanalysis of the Vigenère cipher: The Friedman Test*)

The probability of picking two of the same letters according to the frequency table above would be:

$$(0.082 * 0.082) + (0.015 * 0.015) + (0.028 * 0.028) \dots$$

aa or bb or cc...

Extrapolating this calculation for all pairs of identical letters gives us the Index of Coincidence value of:

$$I = 0.065$$

However, if something like a Vigenère cipher was used then the probability of picking identical letters would be equal as it is simply random letters and not real words. The frequency of letters used above is based on words in the English alphabet which is why it is not used here. Therefore, the Index of Coincidence value for a Vigenère cipher would be:

$$I_v = \left(\frac{1}{26} * \frac{1}{26}\right) + \left(\frac{1}{26} * \frac{1}{26}\right) + \left(\frac{1}{26} * \frac{1}{26}\right) \dots$$

aa or bb or cc...

Extrapolated for all letters part of the English alphabet gives us a value of:

$$I_v = 0.038$$

Therefore, if the Index of Coincidence for a specific ciphertext is closer to 0.038 then it may be using multiple Caesar shifts, however, if it is closer to 0.065 then it is may only be using one Caesar shift. We can graph the frequencies of letters in the ciphertext we were given to showcase that it truly is a Vigenère cipher, the Index of Coincidence for our specific ciphertext will also be useful in future calculations.

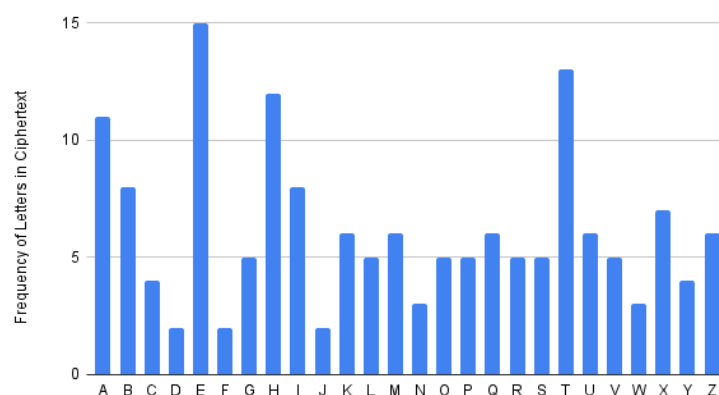


Figure 4

The relatively uniform frequencies disregarding a few outliers suggest a Vigenère cipher has indeed been used. The Index of Coincidence is calculated using the same methods above, by changing each frequency in the graph to a percentage value and doubling each letter. This calculation is not shown to avoid repetitiveness, however, the resultant value is:

$$I_e = 0.0453$$

To find the length of the keyword I will personally attempt to derive a formula based on different cases for certain probabilities using the indices we currently have. First let us assume we know l and we separate all the letters in the ciphertext into:

l number of columns

The value of l does not actually matter and can be virtually any integer, therefore, it will not be showcased using our ciphertext. We can assume that the number of letters in each column is:

$$\frac{n}{l} \text{ letters per column}$$

Similarly to before, let us randomly pick two letters from the ciphertext that are the same. There are two possible cases for this selection:

Case 1: Identical letters are chosen from the same column

Case 2: Identical letters are chosen from different columns

Due to the fact that Vigenère ciphers are typically used with large plaintexts we can assume no column does not have any identical letters. However, it is important to reflect upon this possibility as it would make the cipher virtually impossible to crack if it were much shorter in length.

Consider the situation where a letter has been randomly chosen, the probability of Case 1 occurring for the second letter would be:

$$Probability = \frac{\# \text{ of favourable outcomes}}{\# \text{ of total possible outcomes}}$$

$$P_1 = \frac{\frac{n}{l} - 1}{n - 1}$$

Where the favourable outcome is a second letter being chosen out of all letters in the same column as the original, hence $\frac{n}{l}$, and the total possible outcomes being the all letters in the ciphertext, hence n . In this case, 1 has been subtracted to represent the initial letter that was chosen which cannot be chosen again. Since the letters have been chosen from the same column, it means a singular Caesar shift has been used, therefore, the Index of Coincidence is the same as the English language, therefore the probability of choosing an identical letter is:

$$P_1 = \frac{\frac{n}{l} - 1}{n - 1} * 0.065$$

For Case 2 the probability would be:

$$P_2 = \frac{n - \frac{n}{l}}{n - 1} * 0.038$$

Where the favourable outcome is a letter not part of the group of letters in the initial column, hence $n - \frac{n}{l}$ (all letters - letters in initial column), and the total possible outcomes remaining the same. However, since the letters have been chosen from different columns, this means more than one Caesar shift has been used, therefore the Index of Coincidence is more uniform so the value of 0.038 is used. Since only Case 1 *or* Case 2 can occur we can add the two formulas which is equal to the experimental Index of Coincidence for the ciphertext:

$$I_e = \frac{\frac{n}{l} - 1}{n - 1} * 0.065 + \frac{n - \frac{n}{l}}{n - 1} * 0.038$$

Now we may isolate l as it is our only unknown variable:

$$I_e(n - 1) = (\frac{n}{l} - 1) * 0.065 + (n - \frac{n}{l}) * 0.038$$

$$I_e(n - 1) = \frac{n}{l} * 0.065 - 0.065 + 0.038n - \frac{n}{l} * 0.038$$

$$I_e(n - 1) + 0.065 - 0.038n = \frac{n}{l} (0.065 - 0.038)$$

$$l = \frac{0.027n}{I_e(n-1)+0.065-0.038n}$$

Substituting in our known values for n and I_e and solving for l gives us:

$$l = 3.849$$

Which can be rounded as the key must have an integer length, giving us:

$$l = 4$$

Now that we know the length of the keyword, we can separate our ciphertext into groups of four and conduct a frequency analysis to decipher the message:

***Yulp octu netn deta toui kagk utvh zhxs bchu zeva behw xe.Bw qrlv zaes klhc qthw xara teip
mnhi gtBo mvxa dixk fois myfh zybu etkb yega eiga teih et.Bi qlbl hemo qrxp esht qtap zgkp
shmm arxc qrrv ze.***

Through trial and error showcased above with the Caesar cipher, we can shift letters based on English frequency using the graph in Figure 1 until we get possible words since we know the first letter of every word has been shifted by the same amount, and the second, and so on. In this case, the deciphered message comes out to be:

***Musi ccan beag reat hobb yand itca nhel pcon nect peop le.Ip erso nall ylov etop layt hepi
anob utlh avet ried topl ayma nyin stru ment sint hepa st.Ib elie veth erei ssom ethi ngri ghtf
orev eryo ne.***

Removing the spaces and returning the message to its original state gives us:

Music can be a great hobby and it can help connect people. I personally love to play the piano but I have tried to play many instruments in the past. I believe there is something right for everyone.

Using the table in Figure 2 we can determine the four letter key since we know the plaintext and ciphertext, for example the first letter of the plaintext was **m** which was shifted to y in the ciphertext, the key needed for this would be **m**. Continuously doing so gives the key: **math**

RSA:

Encryption can become even more complicated when the key is made up of numbers rather than a key word or phrase, specifically when two prime numbers are multiplied together. This is the base structure of RSA encryption, named after Ron Rivest, Adi Shamie, and Leonard Adleman. RSA uses the principle idea that multiplying two very large numbers is easy, however factoring such a large product and finding the two original prime numbers would prove to be virtually impossible, which creates a one-way encryption system or a “trapdoor” in the world of cryptography (University of California, Chris Mann, *The Science of Encryption: Prime Numbers and mod N ...*). RSA piqued my interest when I read that it is used to secure emails, it is present in such common day-to-day tools but how does it truly work? To illustrate how RSA encryption works I will create a detailed example using relatively small prime numbers so that the work is easier to follow, however, as highlighted previously it is important to note the strength of RSA

comes from the fact it uses extremely large numbers. The variables and terminology used throughout this example are:

p: Some prime number

q: Some prime number

N: Some integer product

C: Ciphertext value

M: Plaintext/Message value

d: Integer used for deciphering

e: Integer used for enciphering

Private Key: Values kept secret

Public Key: Values that may be shared

The two prime numbers will be p and q which will be kept secret, this will be the private key.

They will then be multiplied together creating a number N where:

$$N = pq$$

N will serve as the public key, due to the fact that even if you know N it will be difficult to find p and q . Suppose there are two people, Person A and Person B who want to use this system to send encrypted messages to each other. Person A will make the public key (N) and Person B will use that key to send Person A a message. For the sake of the example, we can assume they have agreed on a method to encode their message as a number.

Creating the key:

1. Person A will choose two numbers p and q for the private key, suppose they choose:

$$p = 17 \text{ and } q = 19$$

2. Person A multiplies these numbers to create the public key which is given to Person B.

$$N = pq$$

$$N = (17)(19) = 323$$

Person A also chooses another number, e , which is coprime to $(p - 1)(q - 1)$, which means both values must only share a factor of 1. It will also be part of the public key and is given to Person B, it will be used in future calculations.

$$(17 - 1)(19 - 1)$$

$$(16)(18) = 288$$

A value that can be chosen for e is 11, because 11 and 288 only share a factor of 1, it is important to note that e must be less than N .

$$\therefore e = 11$$

Enciphering the message:

1. Person B now has the information necessary for enciphering a message and sending it to Person A, suppose the message (M) is 81.

$$M = 81$$

2. Person B now creates a ciphertext (C) which will be sent to Person A using all of the previous information with the following formula (University of California, Chris Mann, *The Science of Encryption: Prime Numbers and mod N ...*):

$$C = M^e \text{ mod } N$$

$$C = 81^{11} \text{ mod } 323$$

$$C = 984770902183611232881 \bmod 323$$

$$C = 310$$

3. Person B now has their encrypted message which is 310 and they can now send it to Person A.

Deciphering the message:

1. Person A now wants to decode the message they received which is 310. To do so they must find a number, d , such that (University of California, Chris Mann, *The Science of Encryption: Prime Numbers and mod N ...*):

$$ed = 1 \bmod [(p - 1)(q - 1)]$$

This comes from modular inverse, which basically means when two numbers, let us say a and b , are coprime, there exists some integer x making the following formula true:

$$ax = 1 \bmod b$$

In this case, as previously defined, the modulo restricts d to certain possible values as d must be a value such that when it is multiplied by e it is equal to some multiple of:

$$[(p - 1)(q - 1)] + 1$$

Therefore, we can rewrite the equation as:

$$ed = 1 + k[(p - 1)(q - 1)]$$

Where $k \in \mathbb{Z}$ in order to account for possible multiples of $[(p - 1)(q - 1)]$. We can now manipulate the equation to isolate d which gives us the formula:

$$d = \frac{1 + k(\phi N)}{e}$$

$$\text{Where } \phi N = [(p - 1)(q - 1)]$$

After substituting in our known values, we can begin testing whole number values for k until we get an integer answer for d which makes the equation correct.

Table 1: Trial Integers to Find a Corresponding Integer Output

Trial Values of k	Corresponding Value of d
1	$d = \frac{1 + 1(288)}{11}$ $d = 26.3$
2	$d = \frac{1 + 2(288)}{11}$ $d = 52.5$
3	$d = \frac{1 + 3(288)}{11}$ $d = 78.6$
4	$d = \frac{1 + 4(288)}{11}$ $d = 104.8$
5	$d = \frac{1 + 5(288)}{11}$ $d = 131$

Creating this table and testing possible values took about 1 minute and 37 seconds, which means deciphering RSA manually is in fact practical. Through the five trials, we've found the value of d to be 131 and can use this to continue with deciphering the message

2. Person A now has all the necessary values to completely decipher the cipher text (C) they received and can do so using the following formula:

$$M = C^d \bmod N$$

$$M = 310^{131} \bmod 323$$

Such large exponentiation seems very difficult to solve but can be made somewhat easier using binary expansion. Binary expansion is simply splitting the exponent into powers that are multiples of 2 or the power of 1 which add to the original value. Then the products of each calculation can be multiplied together due to basic exponent laws.

$$310^{131} = 310^{128} * 310^2 * 310^1$$

This method can be used continuously to further break down the exponents, however, for the sake of simplicity an approximate scientific value will be used in the following calculations where the original message is recovered:

$$M = 310^{131} \bmod 323$$

$$M \cong 2.3 * 10^{326} \bmod 323$$

$$M = 81$$

How this works:

The mathematics behind RSA rely heavily on Euler's theorem which states (Imperial College London, *Euler's Function & Theorem*):

$$M^{\phi N} \equiv 1 \bmod N$$

For simplicity, let:

$$\phi N = x$$

Using the enciphering algorithm, we can combine it with Euler's theorem to verify whether the deciphering algorithm works for all values and can be consistently used. This will also allow for a deeper understanding of how RSA functions.

$$C = M^e \bmod 323$$

In this verification, “ $\equiv$ ” will mean two values have the same remainder when divided by N . With this we can remove the repeated modulo making the calculations easier to follow.

$$1. C^d \equiv (M^e)^d$$

$$2. C^d \equiv M^{ed}$$

$$3. C^d \equiv M^{ed-1} * M$$

$ed - 1$ is a multiple of x with reference to the previously used formula:

$$ed = 1 + k[(p - 1)(q - 1)]$$

Therefore:

$$ed - 1 = kx$$

Where $k \in \mathbb{Z}$.

$$4. C^d \equiv M^{kx} * M$$

$$5. C^d \equiv (M^x)^k * M$$

Using Euler's Theorems allows us to simplify M^x :

$$6. C^d \equiv 1^k * M$$

Due to the fact that $k \in \mathbb{Z}$:

$$1^k = 1$$

The basic rules of multiplication state by definition that the product of any number multiplied by 1 will be itself, therefore:

$$1 * M = M$$

On this basis, we can come to the conclusion:

$$7. C^d \equiv M$$

∴ The decryption algorithm has been successfully verified and can be reliably used to decipher ciphertexts created from the encryption algorithm.

Conclusion:

The aim of this paper has indeed been achieved, I am confident that my exploration of various ciphers such as RSA and the Vigenère led me to better understand the world of cryptography as a whole. Initially, cryptography seemed like an extremely complex web of encoding and complicated formulas thus I was overwhelmed, however, taking a step-by-step approach throughout this paper and beginning with the simplest ciphers such as the Caesar cipher allowed me to form a basic grasp and fundamental understanding of what cryptography is. From there, I successfully continued to build upon this framework of knowledge by employing my personal examples using certain methods which gradually increased in difficulty alongside performing frequency analysis, graphing, and the derivation and verification of formulas. Cryptography is continuously evolving alongside technology and algorithms previously thought to be unbreakable face threats from our growing industries. RSA for example is estimated to be broken through brute force within the next decade with quantum computing. Future developments like this will lead to increased understanding and the possibility for new ciphers to emerge which I believe is utterly fascinating.

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